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Portfolio Optimization: Evaluating Models for Risk-Adjusted Returns

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Abstract

This paper investigates portfolio optimization strategies applied to 15 Nifty Sectoral Indices over the investment horizon 2023–2025, using historical data from 2010–2023. We compare three approaches: an Equal-Weighted baseline, the classical Markowitz Mean-Variance model, and an enhanced variant termed *Markowitz++*, which incorporates trading costs, leverage limits, and worst-case risk adjustments tuned specifically for Indian equity markets. Through rolling-window backtesting and out-of-sample validation, we demonstrate that Markowitz++ achieves superior risk-adjusted performance across nearly all metrics, attaining a Sharpe ratio of 2.006 and an annualised return of 54.97% in the test period, while maintaining lower volatility than its peers.

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1 Problem Statement

The central objective of this study is to determine the optimal allocation of weights among selected Nifty Sectoral Indices for the investment horizon spanning 2023 to 2025. This optimisation is performed using historical market data from 2010 to 2023, with the goal of maximising expected returns while adhering to a predefined level of risk.

Portfolio optimisation is a classical problem in financial mathematics. The foundational approach is the Markowitz Mean-Variance (MV) model [1], which provides an elegant framework for constructing efficient portfolios. However, the MV model carries a well-known practical limitation: it is highly sensitive to estimation errors in expected returns and covariances. Small perturbations in these inputs can produce drastically different—and often unstable—optimal weight vectors.

This instability has motivated the adoption of simpler heuristics such as the *equal-weighted* portfolio, where capital is spread uniformly across all assets. While theoretically suboptimal, equal-weighting offers robustness and has been shown to be surprisingly competitive in empirical studies.

The challenge therefore extends beyond finding mathematically optimal weights. It involves choosing an optimisation model that appropriately balances theoretical rigour with practical robustness under the following constraints:

- **No short-selling**: weights are constrained to $[0, 1]$.
- **Target risk**: annualised portfolio volatility capped at $\sigma_{\text{tar}} = 17\%$.
- **Transaction costs**: trading activity penalised to reflect realistic bid-ask spreads.
- **Indian market structure**: parameters calibrated to NSE sectoral indices rather than US equities.

In this paper we implement, back-test, and compare three portfolio strategies on Indian equity data, evaluating each on risk-adjusted return, drawdown, volatility, and turnover.

2 Methodology

2.1 Model Development

To evaluate portfolio optimisation strategies under practical market conditions, we construct and compare three portfolio models of increasing complexity: the Equal-Weighted Portfolio, the Basic Markowitz Mean-Variance model, and an enhanced robust optimisation model, Markowitz++.

Equally Weighted Portfolio (EWP). The equally weighted portfolio serves as the baseline benchmark. Under this strategy, capital is distributed equally across all n assets:

$$w_i = \frac{1}{n}, \quad i = 1, 2, \ldots, n$$

This approach requires no estimation of expected returns or covariance matrices, making it simple, robust, and free from estimation error. Although it does not explicitly optimise for risk or return, it provides a useful benchmark for evaluating the effectiveness of optimisation-based strategies.

Basic Markowitz Model. The classical Markowitz Mean-Variance framework aims to maximise expected portfolio return while controlling portfolio risk. The optimisation problem is formulated as:

$$\max_{w} w^\top \mu$$

subject to

$$w^\top \Sigma w \leq \sigma_{\text{tar}}^2, \quad \mathbf{1}^\top w = 1, \quad 0 \leq w_i \leq 1$$

where w denotes the portfolio weight vector, μ is the expected return vector, and Σ is the covariance matrix of returns.

This model generates theoretically efficient portfolios but is known to be highly sensitive to estimation errors in μ and Σ, often resulting in unstable or concentrated allocations.

Markowitz++ Model. To improve robustness, we implement an enhanced Markowitz framework adapted from Boyd et al. [**?**]. This model extends the classical formulation by incorporating penalties for transaction costs, leverage, turnover, and uncertainty-adjusted risk.

These additional constraints and penalty terms help reduce extreme portfolio weights, improve stability, and better reflect realistic trading conditions. The model is adapted for annual rebalancing and calibrated to Indian sectoral indices.

2.2 Backtesting Procedure

To evaluate the practical performance of the three strategies, we use a rolling-window backtesting framework.

Rolling-window framework. A 3-year historical lookback window is used to estimate model parameters, followed by a 1-year out-of-sample test period. This process is repeated sequentially from 2013 to 2023, generating ten independent test windows.

This approach avoids look-ahead bias and allows the models to adapt to changing market conditions over time.

Parameter tuning. The Markowitz++ model introduces multiple hyperparameters controlling penalties for risk, leverage, turnover, and transaction costs. These parameters are selected using grid search across the rolling backtest periods based on performance metrics.

Final optimisation. After selecting the best-performing parameter set, the model is re-estimated using data from 2018–2023 to obtain the optimal portfolio weights for the final investment period 2023–2025.

Performance evaluation. Portfolio performance is compared using the following metrics:

- Sharpe Ratio (risk-adjusted return)
- Cumulative Return
- Annualised Volatility
- Maximum Drawdown
- Portfolio Turnover

These metrics provide a comprehensive comparison of return, risk, and portfolio stability across all three models.

3 Data

3.1 Source

The dataset used in this study consists of daily closing prices of 15 Nifty Sectoral Indices obtained from the National Stock Exchange (NSE) of India [5]. These sectoral indices represent major segments of the Indian equity market, including sectors such as Banking, Information Technology, Automobiles, Pharmaceuticals, Metals, Energy, etc.

The sample period spans from 5 January 2010 to 31 December 2024, covering multiple market cycles, provides sufficient variation in market conditions for both model training and out-of-sample evaluation.

3.2 Preprocessing

Before portfolio construction, the raw price data is transformed into a consistent return dataset suitable for optimisation.

Daily Return Calculation. For each sectoral index i and trading day t, daily simple returns are computed as:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

where $P_{i,t}$ denotes the closing price of asset i on day t.

Return series are used instead of raw prices because portfolio optimisation models depend on expected returns and covariance structures rather than absolute price levels.

Data Alignment and Merging. The individual return series for all 15 indices are aligned by trading date and merged into a single $T \times n$ return matrix, where T is the number of trading days and $n = 15$ is the number of assets.

This unified matrix serves as the primary input for estimating expected returns, covariance matrices, and portfolio weights.

Handling Missing Values. Missing dates arise naturally due to weekends and market holidays. Since these dates are common across all indices and correspond to non-trading periods, they are removed entirely from the dataset.

No interpolation or imputation is performed, ensuring that the return series reflects only actual market activity.

Training and Testing Split. For backtesting purposes, the processed dataset is divided dynamically using a rolling-window structure. At each iteration, a 3-year historical window is used for parameter estimation and portfolio optimisation, followed by a 1-year out-of-sample evaluation period.

4 Optimisation Models

4.1 Basic Markowitz Model

The classical Markowitz Mean–Variance framework seeks to maximise expected portfolio return while controlling portfolio risk under a no-short-selling constraint. The optimisation problem is formulated as:

$$\max_{w} \quad w^\top \mu \tag{1}$$
$$\text{s.t.} \quad w^\top \Sigma w \leq \sigma_{\text{tar}}^2 \tag{2}$$
$$\mathbf{1}^\top w = 1 \tag{3}$$
$$0 \leq w \leq 1 \tag{4}$$

where:

- $w \in \mathbb{R}^n$ denotes the portfolio weight vector,
- $\mu \in \mathbb{R}^n$ is the vector of expected asset returns,
- $\Sigma \in \mathbb{R}^{n \times n}$ is the covariance matrix of returns,
- σ_{tar} is the target annualised volatility.

The objective function is linear, while the risk constraint is quadratic, making this a convex quadratic optimisation problem. We solve it using standard interior-point methods.

4.2 Markowitz++ Model

The basic Markowitz model is highly sensitive to estimation error in expected returns and covariance matrices. To improve robustness, we implement an enhanced optimisation framework based on Boyd et al. [2], incorporating transaction costs, leverage constraints, turnover penalties, and uncertainty-adjusted risk.

The optimisation problem is given by:

$$\begin{aligned} \max_{w,c,z} \quad & \mu^\top w + r_f c - \rho \|w\|_1 - \gamma_{\text{trade}}\, \kappa_{\text{spread}}^\top |z| \\ & - \gamma_{\text{risk}} \max\left(0, \sigma_w + \rho_f \|w\|_1 - \sigma_{\text{target}}\right) \\ & - \gamma_{\text{lev}} \max(0, L - L_{\text{tar}}) - \gamma_{\text{turn}} \max(0, T - T_{\text{tar}}) \end{aligned} \tag{5}$$

subject to:

$$\mathbf{1}^\top w + c = 1 \tag{6}$$

$$z = w - w_{\text{pre}} \tag{7}$$

$$w^{\min} \leq w \leq w^{\max} \tag{8}$$

$$c^{\min} \leq c \leq c^{\max} \tag{9}$$

$$z^{\min} \leq z \leq z^{\max} \tag{10}$$

$$L \leq L_{\text{tar}}, \quad T \leq T_{\text{tar}}, \quad \sigma_w \leq \sigma_{\text{tar}} \tag{11}$$

where:

- $w \in \mathbb{R}^n$ is the vector of risky asset weights,
- $c \in \mathbb{R}$ is the cash allocation,
- $z = w - w_{\text{pre}}$ is the trade vector,
- $L = \|w\|_1$ is the portfolio leverage,
- $T = \frac{1}{2}\|z\|_1$ is the portfolio turnover,
- $\sigma_w = \sqrt{250}\,\|L_{\text{chol}} w\|_2$ is the annualised portfolio volatility.

The Markowitz++ formulation remains convex despite the additional penalty terms, allowing efficient global optimisation using CVXPY [3].

Robustness Advantage. The ℓ_1 regularisation term $\rho\|w\|_1$ discourages extreme portfolio allocations, while the risk penalty term protects against model uncertainty and covariance mis-specification. Together, these additions improve out-of-sample stability and reduce the estimation sensitivity of the classical Markowitz model.

5 Backtest Results

5.1 Rolling-Window Backtest (2013–2023)

To evaluate the performance of the three portfolio strategies under realistic market conditions, we implement a rolling-window backtest using a 3-year training window followed by a 1-year out-of-sample evaluation period. The target annualised portfolio volatility is fixed at 17% across all models to ensure comparability.

For the Markowitz++ framework, hyperparameters controlling risk, turnover, leverage, and transaction cost penalties are selected using grid search over the rolling backtest periods.

The backtesting results are summarised in Table 1.

Table 1: Rolling-window backtesting performance (2013–2023)

Strategy	Sharpe Ratio	Cum. Return	Ann. Volatility	Max Drawdown
Equal-Weighted	0.7330	320.03%	17.11%	0.4157
Basic Markowitz	0.6537	269.21%	17.45%	0.3481
Markowitz++	0.8488	367.03%	15.60%	0.3361

The results show that the Markowitz++ model outperforms both alternative strategies across all major evaluation metrics. It achieves the highest Sharpe ratio, indicating superior risk-adjusted returns, along with the highest cumulative return over the full backtesting period.

Notably, despite operating under the same target volatility constraint, Markowitz++ records the lowest realised volatility and the smallest maximum drawdown, suggesting improved risk control and greater portfolio stability.

In contrast, the classical Markowitz model performs worst in terms of Sharpe ratio and cumulative return. This underperformance reflects its known sensitivity to estimation error, which often leads to unstable and inefficient portfolio allocations.

5.2 Out-of-Sample Test (2023–2025)

After identifying Markowitz++ as the best-performing strategy during backtesting, we re-estimate the model using the most recent training window (2018–2023) and apply the resulting optimal portfolio weights to the live investment period from 2023 to 2025.

This out-of-sample test evaluates whether the superior performance observed during backtesting generalises to unseen market conditions.

The results are presented in Table 2.

Table 2: Out-of-sample performance (2023–2025)

Strategy	Sharpe Ratio	Return	Ann. Volatility	Max Drawdown
Equal-Weighted	1.5110	50.45%	12.74%	0.1033
Basic Markowitz	1.0271	32.68%	12.60%	0.0851
Markowitz++	2.0061	54.97%	10.22%	0.1088

The out-of-sample results are consistent with the backtesting analysis. Markowitz++ achieves the highest Sharpe ratio and total return while maintaining the lowest annualised volatility among all strategies.

Although the basic Markowitz model records the lowest maximum drawdown, this comes at the cost of substantially lower returns and weaker risk-adjusted performance.

Overall, the results indicate that the additional robustness mechanisms in Markowitz++ improve both portfolio efficiency and stability, allowing the model to maintain stronger performance under changing market conditions.

Figure 1 illustrates the cumulative normalised growth of the three portfolios over the investment period. The Markowitz++ portfolio remains ahead for most of the period, with its outperformance becoming particularly visible during strong upward market movements.

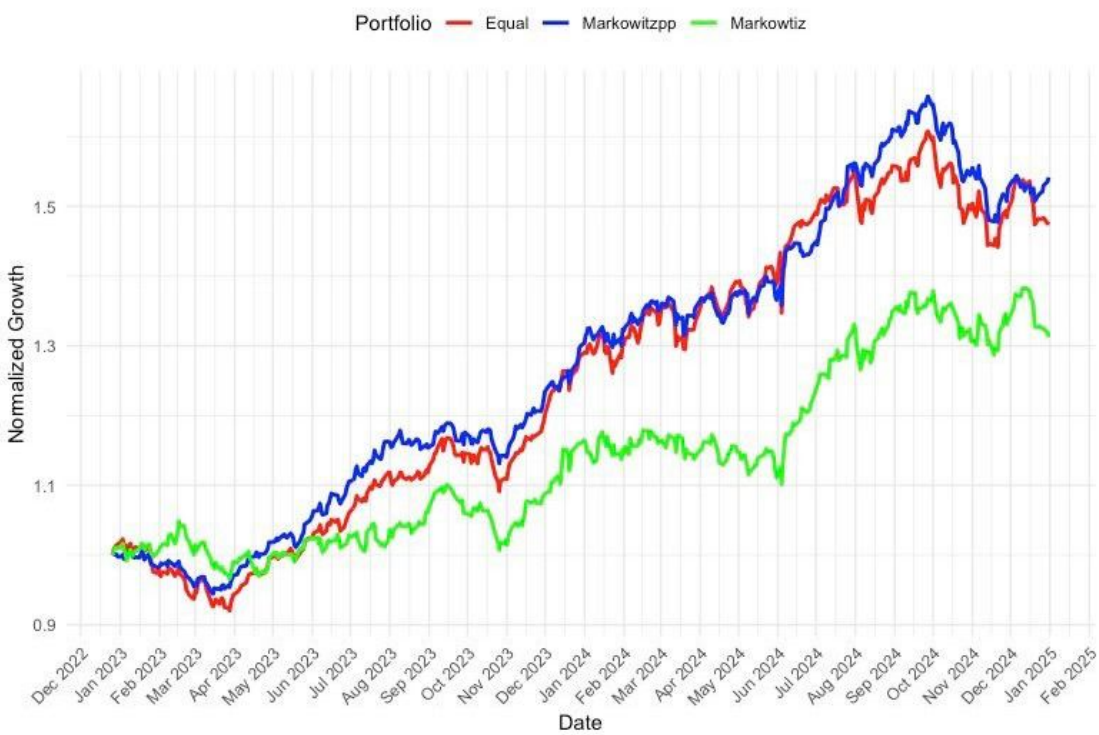

Figure 1: Normalised portfolio growth from 31 December 2022 to 1 February 2025. Markowitz++ consistently outperforms the equal-weighted and classical Markowitz portfolios over most of the investment horizon.

6 Conclusion

This study evaluated three portfolio optimisation strategies on Indian sectoral equity indices over the investment period 2023–2025, using historical market data from 2010–2023 for model estimation, parameter tuning, and backtesting. The objective was to examine whether enhancements to the classical Markowitz framework could improve portfolio stability and risk-adjusted returns under realistic market conditions.

Our results show that the **Markowitz++ model** consistently outperforms both the equally weighted benchmark and the classical Markowitz Mean–Variance model across both rolling-window backtests and out-of-sample evaluation. By incorporating explicit penalties for estimation uncertainty, transaction costs, leverage, and portfolio turnover, the Markowitz++ framework produces allocations that are more stable, better diversified, and more robust to parameter uncertainty.

In the live investment period (2023–2025), the Markowitz++ portfolio achieved the highest Sharpe ratio (2.006) and the highest cumulative return (54.97%), while maintaining lower annualised volatility than the equally weighted portfolio. These results suggest that robustness-enhanced optimisation can significantly improve practical portfolio performance, particularly in emerging market settings such as India where sectoral behaviour can be highly dynamic.

In contrast, the classical Markowitz model consistently underperformed across most evaluation metrics. This reinforces a well-established limitation of mean–variance optimisation: its strong dependence on accurate estimation of expected returns and covariance matrices. In practice, small estimation errors can lead to unstable and overly concentrated portfolios, reducing out-of-sample performance.

Overall, our findings suggest that incorporating practical market frictions and uncertainty-aware regularisation into portfolio optimisation can lead to superior and more reliable investment strategies compared to purely theoretical formulations.

Future extensions of this work may include:

- extending the framework to individual equities or multi-asset portfolios,
- incorporating more realistic transaction cost models using NSE tick-level data,
- exploring dynamic rebalancing frequencies and adaptive volatility targets,
- testing model robustness across different market regimes such as bull, bear, and sideways markets.

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Individual Contributions

Aklanta Borah: implementation of optimization models and backtesting framework in Python; adaptation of the Markowitz++ model; report drafting; visualization and performance analysis.

Rajveer Vora: literature review on portfolio optimisation models; formulation strategy and practical market insights; project planning and LaTeX editing.

Somesh Biswas: data collection and preprocessing from NSE; model analysis and evaluation; report editing and documentation support.

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Appendix: Nomenclature

The following symbols are used throughout the paper.

Symbol	Definition
$\mu \in \mathbb{R}^n$	Expected returns; mean of in-sample daily returns.
r_f	Daily risk-free rate (`risk_free_rate`).
ρ	Return uncertainty scalar; 20th percentile of $\lvert\mu\rvert$ (`rho`).
$\kappa_{\textbf{spread}} \in \mathbb{R}^n$	Half bid-ask spread per asset (`kappa_spread`).
$\gamma_{\textbf{trade}} = 0.2$	Penalty coefficient for trading costs.
$\gamma_{\textbf{risk}} = 0.01$	Penalty coefficient for excess risk.
$\gamma_{\textbf{lev}} = 0.0005$	Penalty coefficient for excess leverage.
$\gamma_{\textbf{turn}} = 0.0025$	Penalty coefficient for excess turnover.
$\rho_f = 0.08$	Risk uncertainty factor (`rho_factor`).
$\sigma_{\textbf{target}} = 0.17$	Target annualised volatility (`risk_target`).
$L_{\textbf{tar}} = 1$	Target leverage (`L_tar`).
$T_{\textbf{tar}} = 20.0$	Target turnover (`T_tar`).
$\sigma_w = \sqrt{250}\,\lVert L_{\textbf{chol}}\, w\rVert_2$	Annualised portfolio risk; L_{chol} is Cholesky factor of $250 \cdot \Sigma$.